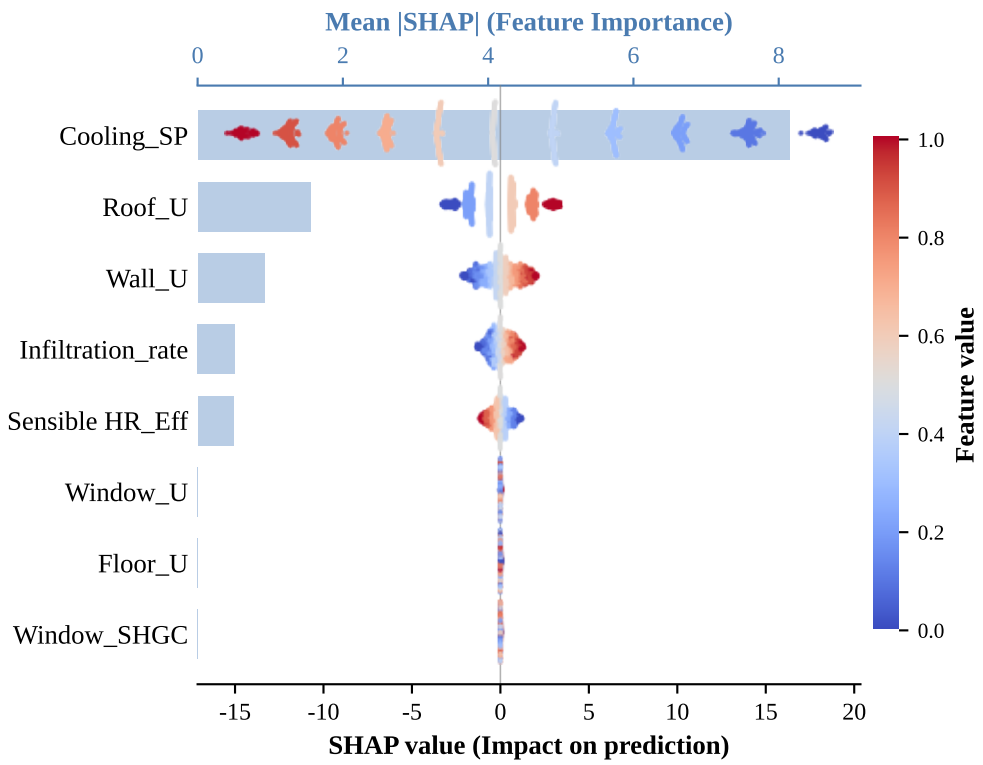
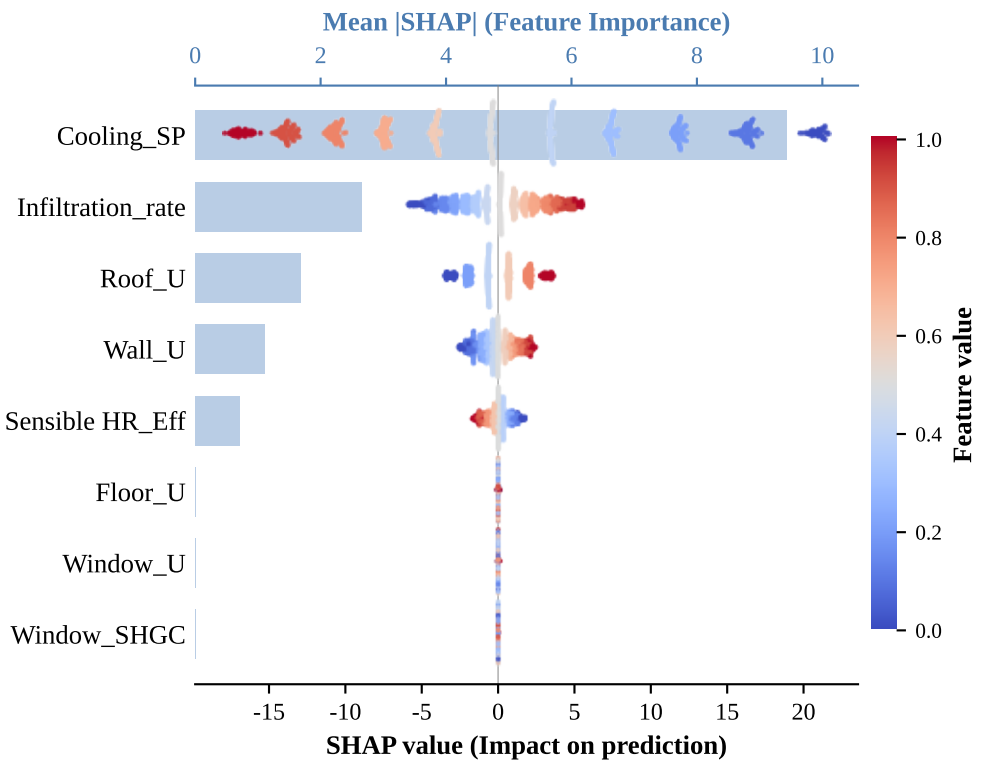
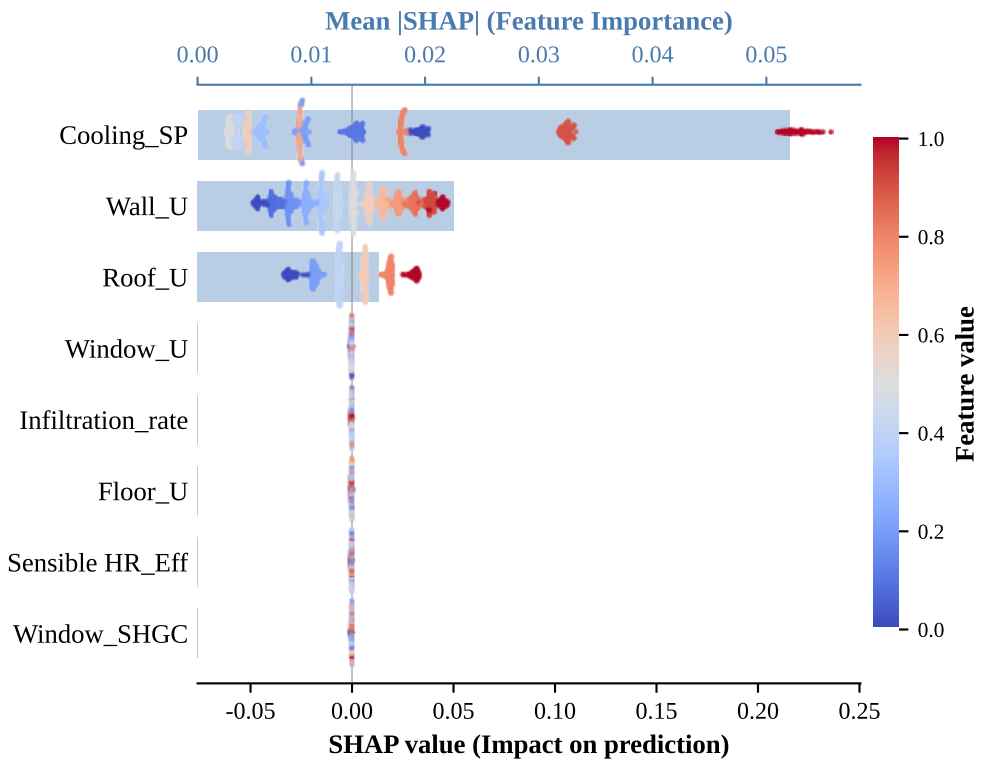
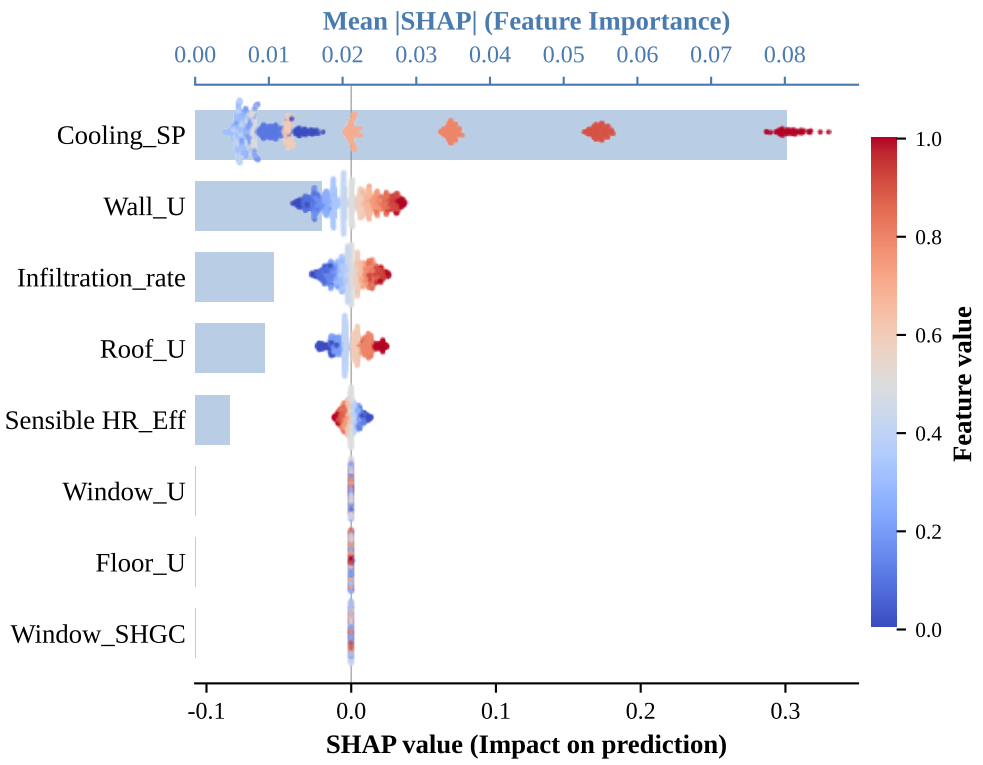
a)


Baseline scenario
Mean |SHAP| (Feature Importance)
0
2
4
6
8
Cooling_SP
Roof_U
Wall_U
Infiltration_rate
Sensible HR_Eff
Window_U
Floor_U
Window_SHGC
1.0
0.8
0.6
0.4
0.2
0.0
Feature value
-15
-10
-5
0
5
10
15
20
SHAP value (Impact on prediction)

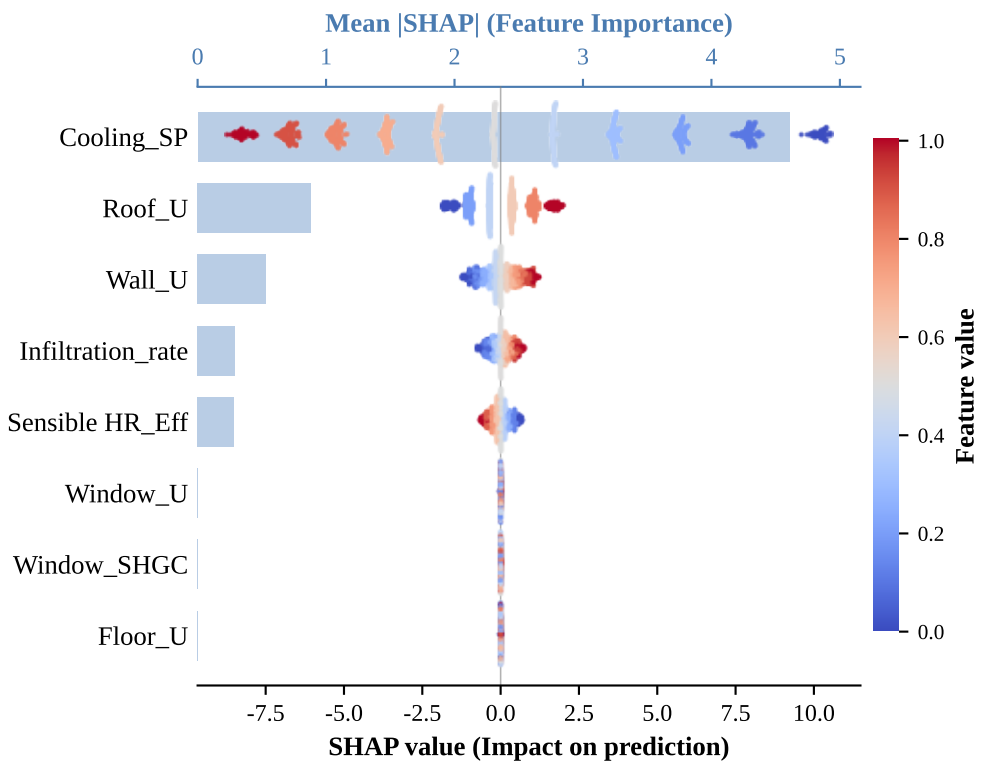

Mid-future scenario
Mean |SHAP| (Feature Importance)
0
2
4
6
8
10
Cooling_SP
Infiltration_rate
Roof_U
Wall_U
Sensible HR_Eff
Floor_U
Window_U
Window_SHGC
1.0
0.8
0.6
0.4
0.2
0.0
Feature value
-15
-10
-5
0
5
10
15
20
SHAP value (Impact on prediction)

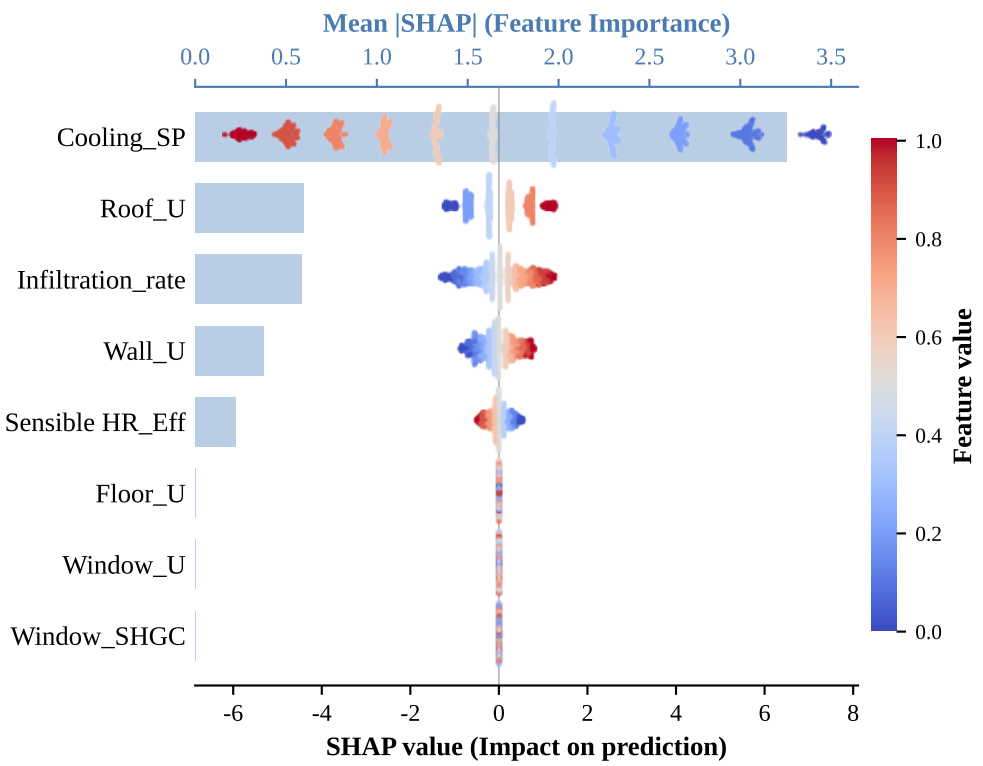
b)


Mean |SHAP| (Feature Importance)
0.00
0.01
0.02
0.03
0.04
0.05
Cooling_SP
Wall_U
Roof_U
Window_U
Infiltration_rate
Floor_U
Sensible HR_Eff
Window_SHGC
1.0
0.8
0.6
0.4
0.2
0.0
Feature value
-0.05
0.00
0.05
0.10
0.15
0.20
0.25
SHAP value (Impact on prediction)


Mean |SHAP| (Feature Importance)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Cooling_SP
Wall_U
Infiltration_rate
Roof_U
Sensible HR_Eff
Window_U
Floor_U
Window_SHGC
1.0
0.8
0.6
0.4
0.2
0.0
Feature value
-0.1
0.0
0.1
0.2
0.3
SHAP value (Impact on prediction)

c)


Mean |SHAP| (Feature Importance)
0
1
2
3
4
5
Cooling_SP
Roof_U
Wall_U
Infiltration_rate
Sensible HR_Eff
Window_U
Window_SHGC
Floor_U
1.0
0.8
0.6
0.4
0.2
0.0
Feature value
-7.5
-5.0
-2.5
0.0
2.5
5.0
7.5
10.0
SHAP value (Impact on prediction)


Mean |SHAP| (Feature Importance)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Cooling_SP
Roof_U
Infiltration_rate
Wall_U
Sensible HR_Eff
Floor_U
Window_U
Window_SHGC
1.0
0.8
0.6
0.4
0.2
0.0
Feature value
-6
-4
-2
0
2
4
6
8
SHAP value (Impact on prediction)